

Strategic AI Agent Development for Texas Hold'em Using Expectiminimax and Monte Carlo Simulation

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Introducing Texas Hold'em

- Texas Hold'em is a popular poker game with hidden information
- Players receive 2 private cards and share 5 community cards
- Goal: make the best 5-card hand or force opponents to fold
- Decisions involve uncertainty, probability, and strategy
- Our project builds an AI agent to play Texas Hold'em intelligently



Project Objective



**Design a strategic
poker-playing AI**



**Handle incomplete
information**



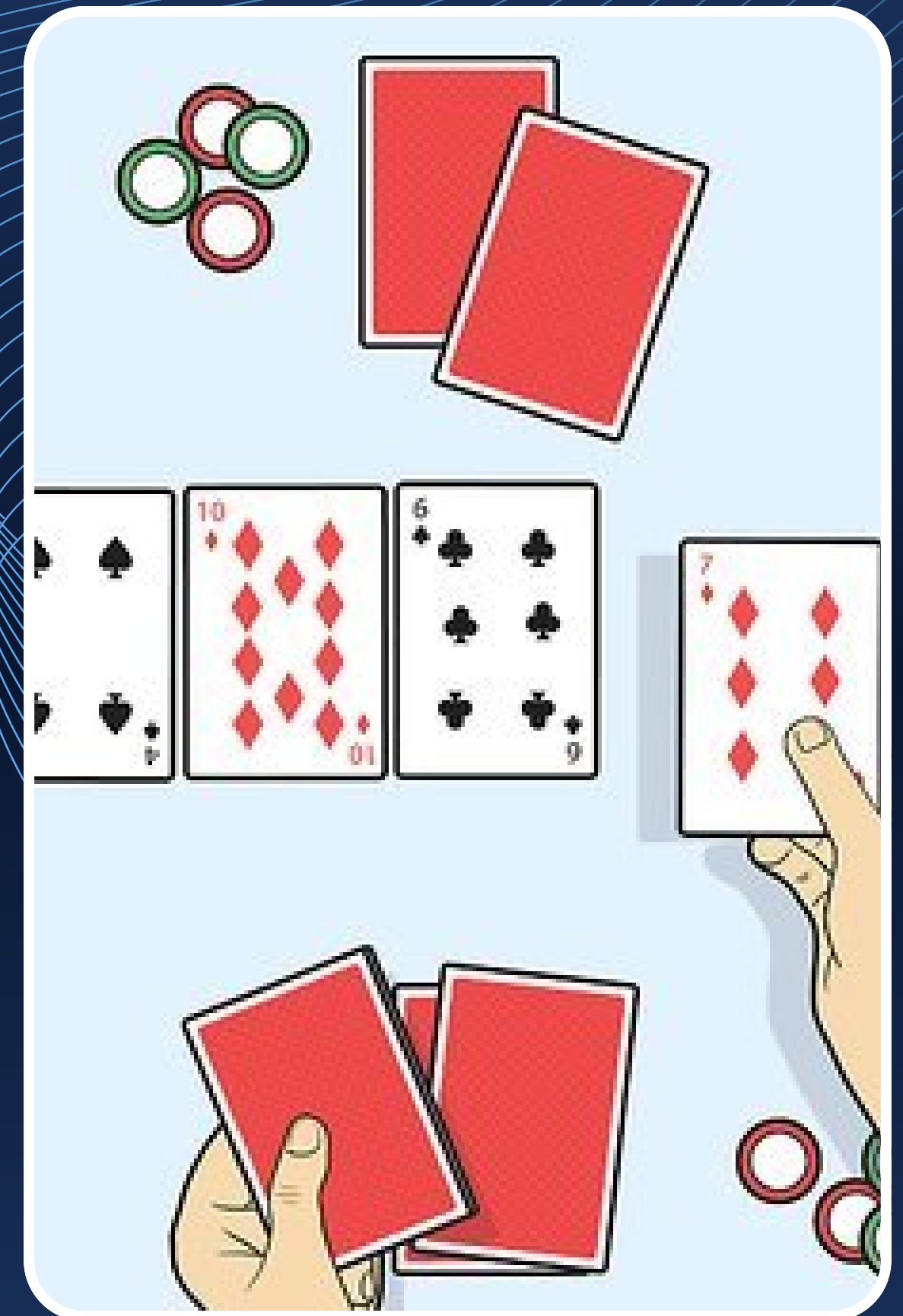
**Make decisions
under random
card dealing**



**Simulate real
human-like poker
reasoning**



**Choose optimal
actions: Fold,
Check, Call, Raise**





Core Methods Used



Monte Carlo Hand Strength Simulation



Balanced Heuristic Evaluation



Expectiminimax Decision Algorithm



Probability + Strategy + Lookahead combined



Estimates win probability
under uncertainty



Randomly simulates:

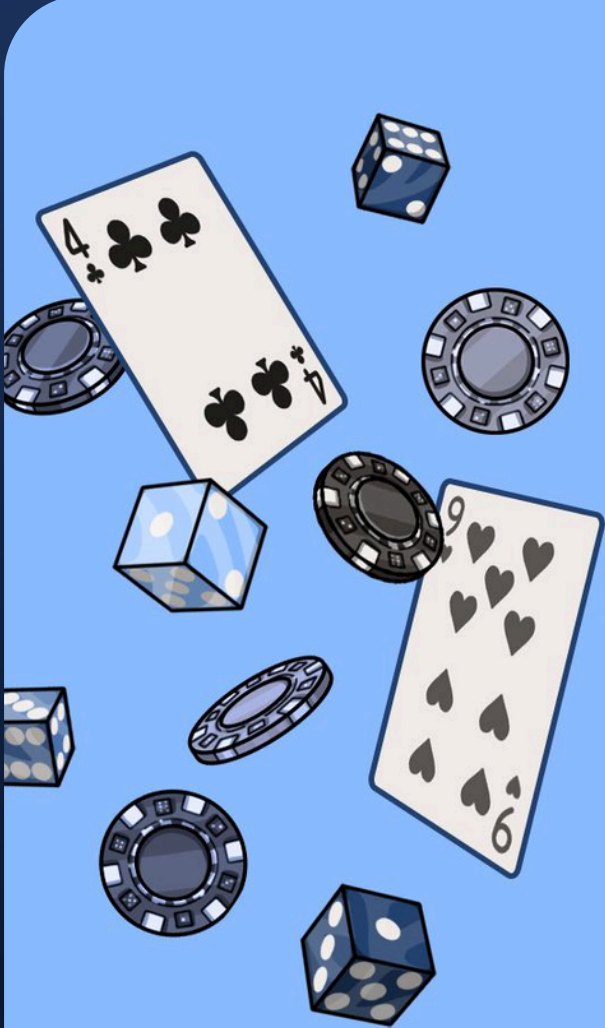
- Opponent cards
- Remaining community cards



Runs hundreds to
thousands of
simulations



Produces realistic hand
strength values. Solves
the problem of unknown
future states



Monte Carlo Simulation

[män-tē 'kär-'lō sim-yə-'lā-shən]

A model used to predict the probability of a variety of outcomes when the potential for random variables is present.

 Investopedia

Monte Carlo Simulation – Significance



**Converts win probability into
a state score**



Encourages:

- Aggressive play for strong hands
- Safe folding for weak hands

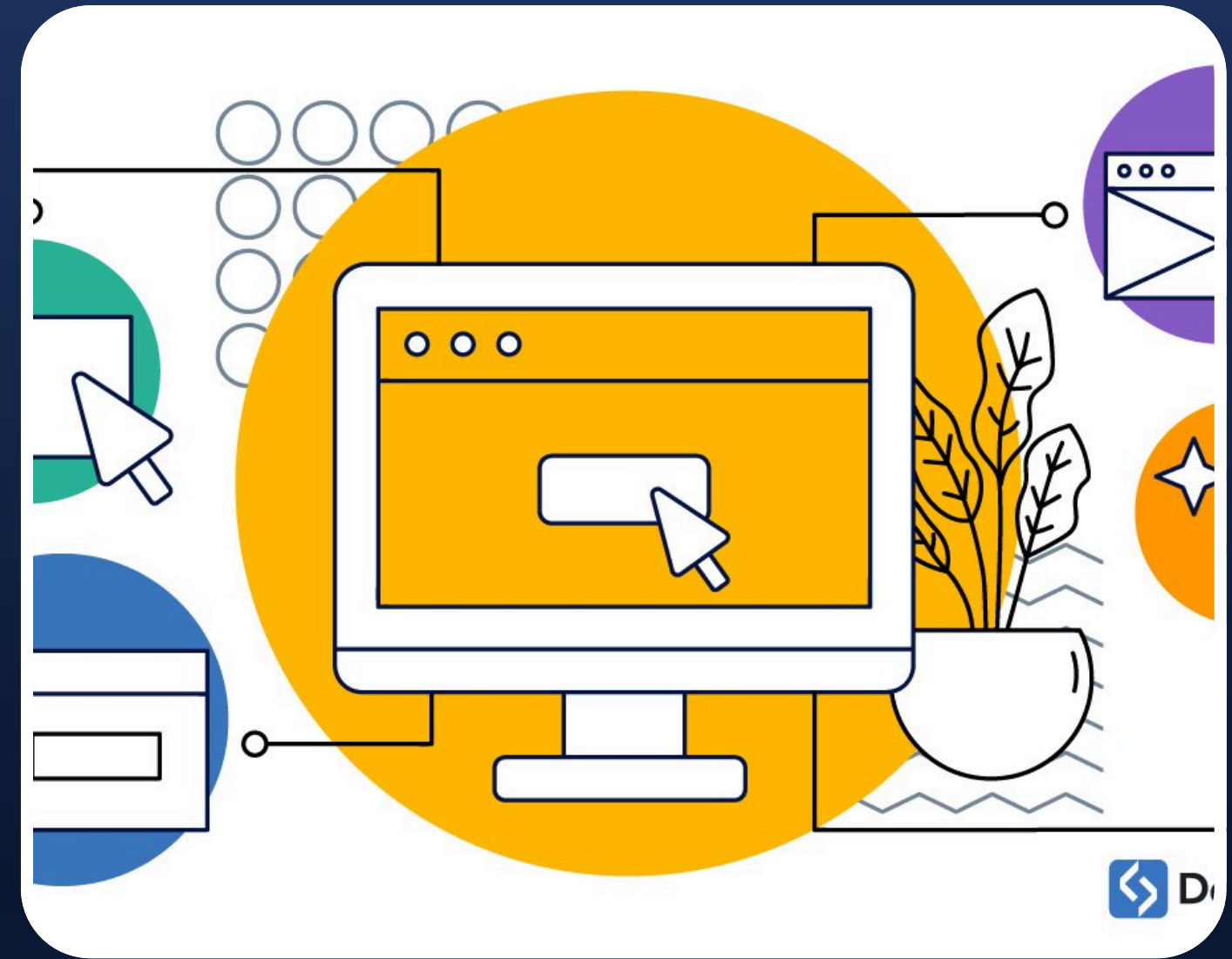


**Considers real poker
factors:**

- Hand strength
- Pot odds
- Calling cost
- Player position
- Stack size



**Makes AI decisions
strategic, not greedy**



Balanced Heuristic Evaluation – Significance



Designed for games with:

- Opponents (MIN nodes)
- AI decisions (MAX nodes)
- Random events (CHANCE nodes)



Searches future states to choose best move

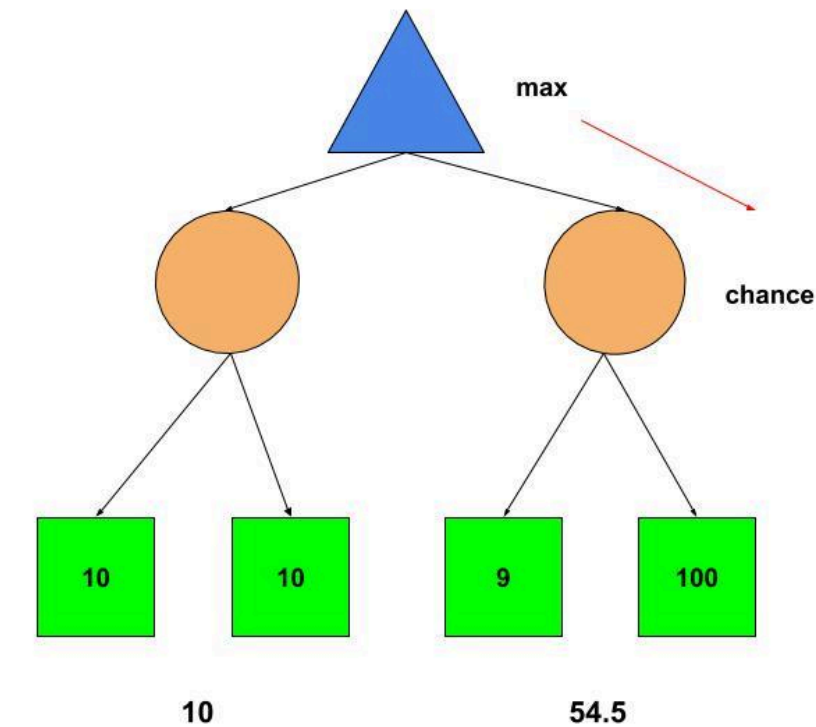


Models real poker flow:

- Betting rounds
- Card dealing
- Opponent actions



Handles uncertainty better than minimax



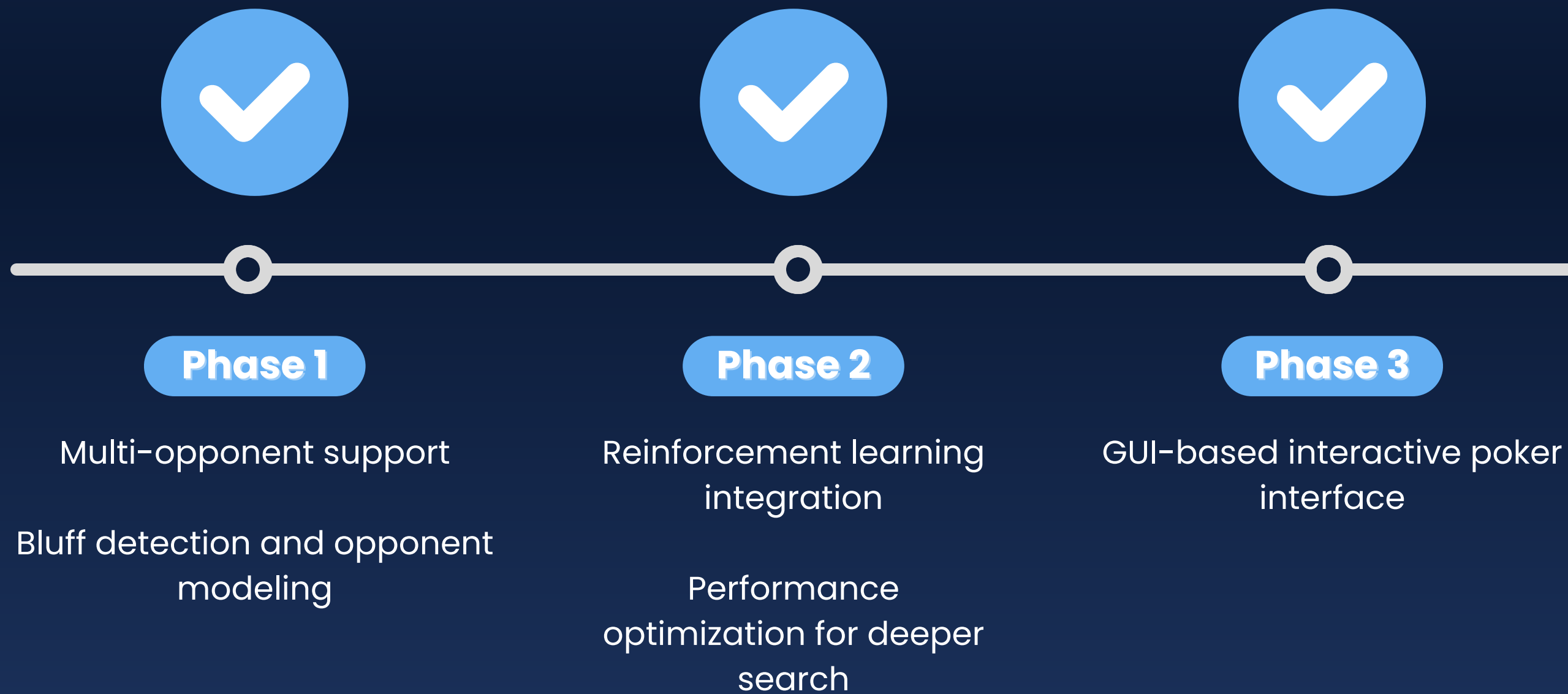
Expectiminimax Algorithm – Significance

Project Significance (Overall)

- ✓ Solves imperfect information problem
- ✓ Mimics real poker decision-making
- ✓ Combines probability + strategy + prediction
- ✓ Produces stable and explainable decisions
- ✓ Suitable for research and AI game modeling



Future Work



Conclusion

- ✓ Built a strategic Texas Hold'em AI
- ✓ Successfully handled uncertainty and randomness
- ✓ Integrated Monte Carlo, heuristic, and expectiminimax
- ✓ AI makes rational and balanced decisions
- ✓ Demonstrates effective AI decision-making in complex games

